

Chemistry
Builds
Better
Thinkers

Small
Steps
Big
Dreams :)

NEET | JEE
Foundation

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Dr. HOMI BHABHA Exam

Practice Questions Set

Class: 9th

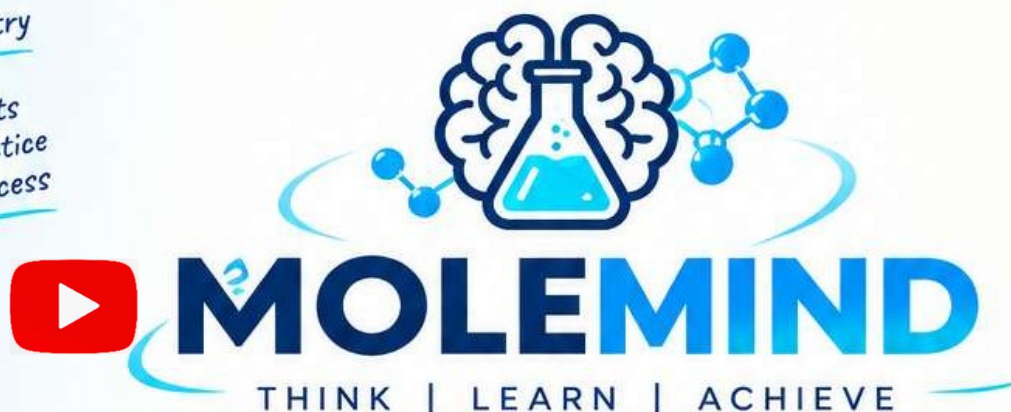
 Isomerism

 Electronic Effects

 Organic Chemistry

 Concepts
→ Practice
→ Success

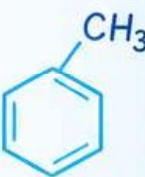

Practice
Learn
Improve
Excel
★



Chapters:

- ✚ Matter, States of Matter & Changes
- ✚ Physical & Chemical Changes
- ✚ Elements, Compounds, Mixtures & Separation Techniques
- ✚ Atoms, Molecules & Structure of Atom
- ✚ Acids, Bases & Indicators

Organic
Chemistry
is not just a subject...
it's a way of thinking
❤



Better
Concepts
Higher
Scores :)

Good
Things
Take
Time ❤


From Confusion to Clarity

— MOLEMIND —



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Practice Question Set

Matter, States of Matter & Changes

Q1. A substance occupies a definite volume but takes the shape of its container. It is most likely:

- A) Solid
- B) Liquid
- C) Gas
- D) Plasma

Q2. Which property of particles best explains why gases can be compressed easily?

- A) They have high mass
- B) They have very strong attractive forces
- C) There are large spaces between particles
- D) Their particles do not move

Q3. When a perfume bottle is opened in one corner of a room, its smell is detected throughout the room after some time. This phenomenon is primarily due to:

- A) Osmosis
- B) Diffusion
- C) Sedimentation
- D) Condensation

Q4. Which change involves absorption of heat from the surroundings?

- A) Freezing of water
- B) Condensation of steam
- C) Melting of ice
- D) Deposition of vapour

Q5. A student observes that ice at 0°C and water at 0°C coexist in a container. During melting, the temperature remains nearly constant because the supplied heat is mainly used for:

- A) Increasing the mass of water
- B) Increasing the kinetic energy of particles
- C) Overcoming intermolecular attractions
- D) Increasing atmospheric pressure

Q6. Which pair represents **two different physical states of the same substance**?

- A) Oxygen and ozone
- B) Water and steam
- C) Carbon and carbon dioxide
- D) Sulphur and sulphur dioxide

Q7. Which observation provides the strongest evidence that matter is made up of moving particles?

- A) A stone has a fixed shape
- B) Sugar dissolves and spreads uniformly in water

- C) Water has a fixed volume
D) Metals are shiny

Q8. A gas is cooled considerably. Which change is most likely?

- A) Particle motion increases
B) Interparticle distance generally decreases
C) Particle mass decreases
D) Number of particles decreases

Q9. Which process is **not** a change of state?

- A) Melting
B) Sublimation
C) Burning
D) Condensation

Q10. Naphthalene balls gradually disappear when left in a cupboard without forming a liquid. This is:

- A) Evaporation
B) Condensation
C) Sublimation
D) Freezing

Conceptual Traps:

Q11. Two identical amounts of water are placed in two vessels. Vessel A has a wide surface and vessel B has a narrow surface. Under otherwise identical conditions, water evaporates faster from:

- A) A
B) B
C) Both at the same rate
D) Cannot be predicted

Q12. Evaporation causes cooling because:

- A) All particles lose energy
B) Higher-energy particles escape from the surface
C) Lower-energy particles escape first
D) The liquid gains heat from the surroundings

Q13. Which condition would generally **decrease** the rate of evaporation?

- A) Increasing temperature
B) Increasing surface area
C) Increasing humidity
D) Increasing wind speed

Q14. Steam at 100°C can cause more severe burns than water at 100°C mainly because steam:

- A) Has no mass
B) Contains latent heat that is released during condensation
C) Has a lower temperature
D) Is chemically different from water



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Q15. A wet cloth dries faster on a windy day because wind:

- A) Decreases the kinetic energy of water molecules
- B) Removes water vapour near the cloth
- C) Decreases the surface area
- D) Converts water directly into ice

Q16. Which statement about boiling is correct?

- A) It occurs only at the surface
- B) It can occur throughout the liquid
- C) It occurs at every temperature
- D) It does not involve energy absorption

Q17. At high altitude, water generally boils at a temperature:

- A) Higher than 100°C
- B) Exactly 100°C
- C) Lower than 100°C
- D) Always 0°C

Q18. If atmospheric pressure increases, the boiling point of a liquid generally:

- A) Decreases
- B) Increases
- C) Becomes zero
- D) Remains unchanged

Q19. Which phenomenon demonstrates that particles possess kinetic energy even when a substance appears stationary?

- A) Diffusion
- B) Filtration
- C) Sedimentation
- D) Decantation

Q20. A solid changes directly into a gas. The reverse process is:

- A) Evaporation
- B) Condensation
- C) Deposition
- D) Fusion

Physical & Chemical Changes

Q21. Which is a **chemical change**?

- A) Melting wax
- B) Cutting paper
- C) Rusting iron
- D) Dissolving sugar

Q22. Which observation provides the strongest evidence of a chemical reaction?

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- A) Change in shape
 - B) Change in size
 - C) Formation of a new substance
 - D) Change in physical state

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Q23. Burning magnesium ribbon produces a white powder. The best conclusion is:

- A) Only a physical change occurred
- B) Magnesium changed into a new substance
- C) Magnesium only changed its shape
- D) No reaction occurred

Q24. Which change is reversible under ordinary conditions?

- A) Burning wood
- B) Rusting iron
- C) Melting ice
- D) Cooking food

Q25. A student says, "Dissolving salt in water is definitely a chemical change because salt disappears." Which is the best response?

- Electronic Effects
- A) Correct, because disappearance always means reaction
 - B) Incorrect, because salt can be recovered by evaporation
 - C) Correct, because water reacts with salt
 - D) Incorrect, because salt is destroyed

Q26. Which combination contains **only physical changes**?

- Organic Chemistry
- A) Melting ice, cutting paper, dissolving sugar
 - B) Burning coal, rusting iron, cooking food
 - C) Curdling milk, burning paper, melting wax
 - D) Digestion, respiration, freezing water

Q27. Formation of curd from milk is considered a chemical change mainly because:

- Concepts Practice Success
- A) Its shape changes
 - B) A new substance with different properties forms
 - C) Its temperature changes
 - D) It becomes thicker only

Q28. Crystallisation of copper sulphate from its solution is generally classified as:

- A) Chemical change
- B) Physical change
- C) Nuclear change
- D) Combustion

Q29. Which process involves both physical and chemical changes?

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- A) Burning a candle
B) Melting ice
C) Breaking glass
D) Freezing water

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Q30. When a candle burns, the wax near the flame first melts and then the vapour burns. Therefore:

- A) Entire process is physical
B) Entire process is chemical
C) Both physical and chemical changes occur
D) No change occurs

Scientific Reasoning

Q31. A student heats a liquid and observes bubbles throughout the liquid. Which observation would best distinguish boiling from evaporation?

- Isomerism
- A) It occurs at the surface
B) It occurs throughout the liquid
C) It causes cooling
D) It involves liquid particles

Q32. Four samples are given:

- Electronic Effects
- P: Ice
 - Q: Water
 - R: Steam
 - S: Liquid nitrogen
- Organic Chemistry

Which pair contains substances in the **same physical state**?

- Concepts
Practice
Success
- A) P and Q
B) Q and R
C) P and S
D) R and S

Q33. A sealed container contains water and water vapour at equilibrium. If some vapour condenses, then eventually:

- A) No evaporation can occur
B) Evaporation can continue until equilibrium is restored
C) All water becomes vapour
D) The container becomes empty

Q34. Which statement is **incorrect**?

- A) Gases exert pressure
B) Liquids have definite volume
C) Solids have no intermolecular forces
D) Gas particles are in continuous motion

Q35. A liquid evaporates faster when its surface area is increased. This suggests that evaporation:



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- A) Depends on the number of particles at the surface
 - B) Depends only on mass
 - C) Occurs only at the bottom
 - D) Is independent of temperature

Olympiad-Level Questions

Q36. A sealed bottle is filled completely with water and kept in a freezer. The bottle may crack because:

- A) Water contracts on freezing
- B) Water expands on freezing
- C) Ice has no volume
- D) Pressure inside becomes zero

Q37. Which statement best explains why a wooden block can float on water while an iron nail sinks?

- Isomerism
- A) Wood contains no particles
 - B) Density and buoyancy are involved
 - C) Iron has no mass
 - D) Water attracts wood chemically

Q38. A drop of ink is added to two beakers containing equal amounts of water. One is at 20°C and the other at 60°C. The ink spreads faster in the hotter water because:

- Electronic Effects
- A) Water molecules become heavier
 - B) Molecular motion increases with temperature
 - C) Ink becomes a gas
 - D) Water stops attracting ink

Q39. A student claims: "When water boils, the molecules themselves become hotter and therefore turn into steam." Which correction is most appropriate?

- Organic Chemistry
- A) Molecules disappear
 - B) Liquid water changes state as energy enables particles to overcome intermolecular attraction
 - C) Water changes into a new chemical substance
 - D) Steam contains different atoms

Q40. Which sequence represents increasing particle freedom of movement?

- A) Gas → Liquid → Solid
- B) Solid → Liquid → Gas
- C) Liquid → Solid → Gas
- D) Solid → Gas → Liquid

Q41. A wet hand feels colder when exposed to moving air because:

- A) Air freezes the water
- B) Evaporation is enhanced and removes heat
- C) Water gains heat
- D) Air increases humidity

Q42. Which situation is most directly explained by **diffusion**?

- A) A stone sinking in water
- B) Smell of incense spreading through a room

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- C) Salt settling at the bottom
D) Water freezing

Q43. If the kinetic energy of particles in a substance increases sufficiently, the substance may:

- A) Change its state
B) Lose all particles
C) Become massless
D) Stop occupying space

Q44. A substance changes from solid $\rightarrow$ liquid $\rightarrow$ gas. Which statement is correct?

- A) Interparticle separation generally increases
B) Particle mass continuously decreases
C) Chemical identity must change at every stage
D) Number of atoms necessarily decreases

Q45. Which change would require **maximum energy input** under suitable conditions?

- A) Freezing
B) Condensation
C) Melting
D) Deposition

Challenge Round

Q46. A student places equal quantities of water in four conditions:

- I. Wide open vessel
II. Narrow open vessel
III. Wide vessel with a fan
IV. Wide vessel in a humid room

Which is expected to evaporate fastest?

- A) I
B) II
C) III
D) IV

Q47. A substance has a fixed volume but no fixed shape. Its particles are:

- A) Completely fixed in position
B) Close together but able to move past one another
C) Extremely far apart with negligible attraction
D) Completely motionless

Q48. Which pair represents **opposite changes of state**?

- A) Melting and freezing
B) Evaporation and sublimation
C) Boiling and melting
D) Condensation and freezing

Q49. A liquid is cooled below its freezing point but remains liquid temporarily. This phenomenon is called:

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- A) Superheating
 - B) Supercooling
 - C) Sublimation
 - D) Deposition

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Q50. Three statements are given:

P: Evaporation can occur below the boiling point.

Q: Boiling occurs only at the surface of a liquid.

R: Increasing wind speed generally increases evaporation.

Which option is correct?

- A) P only
- B) Q only
- C) P and R
- D) Q and R

Elements, Compounds, Mixtures & Separation Techniques

Q51. Which of the following is a **pure substance**?

- Electronic
Effects
- A) Air
 - B) Distilled water
 - C) Brass
 - D) Soil

Q52. Which statement correctly distinguishes a compound from a mixture?

- Organic
Chemistry
- A) A compound always contains two or more elements physically mixed
 - B) Components of a compound can always be separated by physical methods
 - C) A compound has properties different from those of its constituent elements
 - D) A mixture always has a fixed composition

Q53. Which of the following is a **homogeneous mixture**?

- Concepts
Practice
Success
- A) Sand + water
 - B) Oil + water
 - C) Salt solution
 - D) Soil

Q54. Which is an example of a **heterogeneous mixture**?

- A) Air
- B) Brass
- C) Sugar solution
- D) Granite

Q55. A sample contains only hydrogen and oxygen atoms in a fixed ratio of 2:1. The sample is most likely:

- A) A mixture of hydrogen and oxygen
- B) Water
- C) Oxygen dissolved in hydrogen
- D) Hydrogen peroxide

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Q56. Which statement about a mixture is correct?

- A) Its components always combine chemically
 - B) Its composition is always fixed
 - C) Its components retain their individual properties
 - D) It always has a new chemical formula
- Small Steps Big Dreams :)

Q57. Which pair contains **only compounds**?

- A) Water and carbon dioxide
- B) Oxygen and nitrogen
- C) Brass and steel
- D) Air and water

Q58. Which pair contains **only mixtures**?

- A) Air and brass
 - B) Water and carbon dioxide
 - C) Oxygen and nitrogen
 - D) Sodium chloride and air
- Isomerism

Q59. A substance cannot be broken down into simpler substances by ordinary chemical methods. It is classified as;

- A) Mixture
 - B) Compound
 - C) Element
 - D) Solution
- Electronic Effects

Q60. Which of the following is **not** a characteristic of a pure compound?

- A) Fixed composition
 - B) Definite properties
 - C) Components retain all their original properties
 - D) Chemical combination of elements
- Organic Chemistry

Conceptual Traps

Concepts Practice Success

Q61. A mixture contains iron filings, sand and common salt. Which sequence would be most appropriate to separate all three?

- A) Filtration → magnet → evaporation
- B) Magnet → dissolve in water → filtration → evaporation
- C) Evaporation → magnet → filtration
- D) Filtration → evaporation → magnet

Q62. Why is a magnet used first in the separation of iron, sand and salt?

- A) Salt is magnetic
- B) Iron can be selectively removed without affecting sand and salt
- C) Sand becomes magnetic in water
- D) Salt reacts with the magnet

Q63. A mixture contains ammonium chloride and sand. The most suitable method for separating ammonium chloride is:

- Chemistry
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Thinkers
- A) Filtration
 - B) Sublimation
 - C) Decantation
 - D) Magnetic separation

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Q64. A mixture of two liquids has boiling points that differ considerably. The most suitable method is:

- A) Filtration
- B) Sublimation
- C) Simple distillation
- D) Magnetic separation

Q65. Fractional distillation is particularly useful when:

- A) Two solids have different colours
- B) Two liquids have close boiling points
- C) One solid is magnetic
- D) One substance is insoluble in water

Q66. Which method is best for separating **oil and water**?

- Electronic
Effects
- A) Filtration
 - B) Separating funnel
 - C) Sublimation
 - D) Crystallisation

Q67. A student wants to obtain pure water from salt solution. The most appropriate method is:

- Organic
Chemistry
- A) Filtration
 - B) Sedimentation
 - C) Distillation
 - D) Magnetic separation

Q68. During distillation of salt solution, which substance is collected as the distillate?

- Concepts
→ Practice
→ Success
- A) Salt
 - B) Water
 - C) Both salt and water
 - D) Neither

Q69. Why cannot ordinary filtration separate dissolved salt from water?

- A) Salt particles are magnetic
- B) Dissolved salt particles pass through the filter along with water
- C) Water cannot pass through filter paper
- D) Salt evaporates instantly

Q70. Crystallisation is preferred over simple evaporation when the objective is to:

- A) Obtain a purer solid
- B) Obtain a liquid
- C) Separate magnetic substances
- D) Separate immiscible liquids

Experimental & Observation-Based

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Q71. A student adds sand to water and stirs it. After some time, sand settles at the bottom. This process is:

- A) Filtration
- B) Sedimentation
- C) Distillation
- D) Sublimation

Q72. After sedimentation, the clear upper liquid is carefully poured into another vessel. This process is:

- A) Decantation
- B) Crystallisation
- C) Sublimation
- D) Centrifugation

Q73. A muddy water sample is first allowed to stand and then filtered. The purpose of sedimentation before filtration is mainly to:

- Isomerism
- A) Increase the amount of dissolved material
 - B) Remove larger suspended particles before filtration
 - C) Convert the mixture into a compound
 - D) Evaporate the water

Electronic
Effects

Q74. A student uses chromatography to separate the colours in black ink. The separation occurs because different components:

- A) Have identical solubilities
 - B) Move at different rates through the medium
 - C) Have identical masses
 - D) React chemically with paper
- Organic
Chemistry

Q75. In paper chromatography, a component that travels farther generally has:

- A) Greater attraction to the stationary phase
 - B) Greater tendency to move with the solvent
 - C) Zero solubility in solvent
 - D) Greater density only
- Concepts
Practice
Process

Q76. A mixture contains two solids: one soluble in water and the other insoluble. Which sequence is appropriate?

- A) Add water → filter → evaporate filtrate
- B) Filter → add water → evaporate residue
- C) Sublime → filter → distil
- D) Distil → filter → evaporate

Q77. A student heats a mixture containing a substance that sublimates and another substance that does not. The sublimable substance can be collected:

- A) At the bottom of the vessel
- B) On a cooler surface
- C) In the filter paper
- D) In the water layer

Q78. Which technique is based mainly on differences in **particle size**?

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- A) Filtration
 - B) Distillation
 - C) Chromatography
 - D) Sublimation

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Q79. Which technique is based mainly on differences in **boiling points**?

- A) Filtration
- B) Distillation
- C) Magnetic separation
- D) Sedimentation

Q80. Which technique depends on a substance changing directly from solid to vapour?

- A) Distillation
- B) Sublimation
- C) Decantation
- D) Filtration

Scientific Reasoning

Q81. A student says, "Air is a compound because it contains oxygen and nitrogen." Which statement correctly identifies the error?

- A) Air contains only oxygen
- B) Oxygen and nitrogen are chemically bonded in air
- C) Air is a mixture because its gases are not chemically combined
- D) Air is an element

Q82. Which observation most strongly suggests that a sample is a mixture rather than a compound?

- A) It has a definite chemical formula
- B) Its components can be separated physically
- C) It has a fixed composition
- D) It has completely new properties

Q83. Two colourless liquids are mixed. No gas, precipitate or temperature change is observed, and the mixture can later be separated by distillation. The best conclusion is:

- A) They must have formed a compound
- B) They are likely forming a physical mixture
- C) A chemical reaction definitely occurred
- D) Both liquids disappeared

Q84. A sample contains particles of different sizes that can be separated by filtration. The sample is most likely:

- A) A homogeneous compound
- B) A heterogeneous mixture
- C) An element
- D) A pure substance

Q85. Which statement is true for both a compound and a mixture?

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- A) Both always have variable composition
 - B) Both contain more than one kind of atom or substance
 - C) Both can always be separated physically
 - D) Both retain all properties of their components

Olympiad-Level

Q86. A mixture contains salt, sand and ammonium chloride. Which sequence can separate all three components most effectively?

- A) Filtration → evaporation → sublimation
- B) Sublimation → dissolve in water → filtration → evaporation
- C) Dissolve → sublimation → filtration → evaporation
- D) Evaporation → sublimation → filtration

Q87. A student accidentally adds iron filings to a mixture of salt and sand. Which property provides the simplest way to separate iron?

- Isomerism
- A) Solubility
 - B) Magnetism
 - C) Density
 - D) Boiling point

Q88. A mixture contains liquids A and B. A boils at 78°C and B at 100°C . Under suitable conditions, the first major fraction obtained during simple distillation will be richer in:

- Electronic Effects
- A) B
 - B) A
 - C) Both equally
 - D) Neither

Q89. Why is a thermometer bulb placed near the side arm of a distillation flask?

- Organic Chemistry
- A) To measure the temperature of the incoming water
 - B) To measure the temperature of vapour leaving the flask
 - C) To increase boiling point
 - D) To prevent condensation

Q90. A student tries to separate alcohol and water using filtration. Why does this fail?

- Concepts
→ Practice
→ Success
- A) Both are solids
 - B) Both are gases
 - C) They form a homogeneous liquid mixture and pass through filter paper
 - D) Alcohol is magnetic

High-Level Homi Bhabha Challenge

Q91. Consider the following:

1. Brass
2. Distilled water
3. Air
4. Sodium chloride

5. Soil

Which are mixtures?

- A) 1, 3 and 5
- B) 2 and 4
- C) 1, 2 and 3
- D) 3 and 4

Q92. Which pair is incorrectly matched?

- A) Filtration — insoluble solid + liquid
- B) Distillation — liquids with different boiling points
- C) Sublimation — sublimable solid + non-sublimable solid
- D) Chromatography — magnetic substances

Q93. A student has a mixture of **kerosene, water and dissolved salt**. Which method would be most logical?

- A) Separating funnel → distillation
- B) Filtration → sublimation
- C) Sublimation → filtration
- D) Crystallisation → magnetism

Q94. A pure substance undergoes a change and produces two substances with entirely different properties. The original substance was most likely:

- A) An element
- B) A compound
- C) A mixture
- D) A solution

Q95. Which statement is **most scientifically accurate**?

- A) Every homogeneous substance is a compound
- B) Every heterogeneous substance is a mixture
- C) Every mixture is heterogeneous
- D) Every pure substance contains only one atom

Q96. A student heats a salt solution until all water is removed. The remaining salt is recovered. This demonstrates that:

- A) Salt chemically decomposes in water
- B) Dissolution can be reversed by a physical process
- C) Salt becomes a new compound
- D) Water and salt form a permanent compound

Q97. A mixture of two solids has the following properties:

- Solid P is magnetic.
- Solid Q dissolves in water.
- Both are insoluble in kerosene.

Which method is most appropriate?

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- A) Add water → filter → magnet
B) Magnet → add water → filter
C) Add kerosene → filter → evaporate
D) Distillation → magnet

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Q98. A student claims: "If two substances are mixed and no visible change occurs, they must form a mixture." Why is this conclusion not always scientifically sufficient?

- A) A chemical reaction can sometimes occur without an obvious immediate visible change
B) All chemical reactions produce bubbles
C) Mixtures always change colour
D) Compounds can never be colourless

Q99. A sample contains only one type of molecule, and each molecule contains atoms of two different elements chemically bonded in a fixed ratio. The sample is:

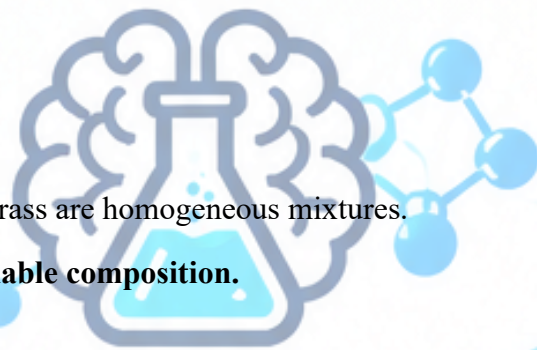
- A) Element
B) Compound
C) Homogeneous mixture
D) Heterogeneous mixture

Q100. A mixture contains **iron filings, ammonium chloride, sand and common salt**. Which sequence gives the most logically complete separation?

- A) Magnet → sublimation → add water → filtration → evaporation
B) Sublimation → magnet → add water → evaporation → filtration
C) Add water → filtration → magnet → sublimation
D) Magnet → filtration → sublimation → distillation

🔥 Important Traps from this set

- **Mixture ≠ always heterogeneous** → air, salt solution and brass are homogeneous mixtures.
- **Compound has fixed composition; mixture can have variable composition.**
- **Dissolving ≠ necessarily chemical reaction.**
- **Filtration separates insoluble solids**, not dissolved substances.
- **Simple distillation → appreciably different boiling points.**
- **Fractional distillation → close boiling points.**
- **Sublimation → solid directly → vapour.**
- **Chromatography → different components travel at different rates.**
- **Separation method depends on a physical property** such as magnetism, solubility, particle size or boiling point.



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Atoms, Molecules & Structure of Atom

Q101. The smallest particle of an element that retains the chemical properties of that element is:

- A) Molecule
- B) Atom
- C) Ion
- D) Nucleus

Q102. Which statement about an atom is correct?

- A) It always exists independently
- B) It is always electrically charged
- C) It is the basic unit of an element
- D) It contains only electrons

Q103. Which subatomic particle has a negative charge?

- A) Proton
- B) Neutron
- C) Electron
- D) Nucleus

Q104. The mass of an atom is concentrated mainly in its:

- A) Electron cloud
- B) Nucleus
- C) Outer shell
- D) Valence shell

Q105. Which particle determines the identity of an element?

- A) Electron
- B) Neutron
- C) Proton
- D) Molecule

Q106. An atom has 11 protons. Its atomic number is:

- A) 10
- B) 11
- C) 12
- D) 22

Q107. An electrically neutral atom contains 17 protons. Number of electrons is:

- A) 16
- B) 17
- C) 18
- D) 34

Q108. An atom has 12 protons and 12 neutrons. Its mass number is:

- A) 12
- B) 24



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- C) 6
D) 0

Q109. An atom has mass number 23 and atomic number 11. Number of neutrons is:

- A) 11
B) 12
C) 23
D) 34

Q110. Isotopes are atoms having:

- A) Same mass number, different atomic number
B) Same atomic number, different mass number
C) Different atomic number and same neutrons
D) Same number of electrons only

Conceptual

Q111. Which pair represents isotopes?

- A) ^{12}C and ^{14}C
B) ^{14}C and ^{14}N
C) ^{23}Na and ^{24}Mg
D) ^1H and ^4He

Q112. Isotopes of an element generally have similar chemical properties because they have the same:

- A) Number of neutrons
B) Mass number
C) Electronic configuration
D) Atomic mass

Q113. Which subatomic particle contributes approximately one atomic mass unit?

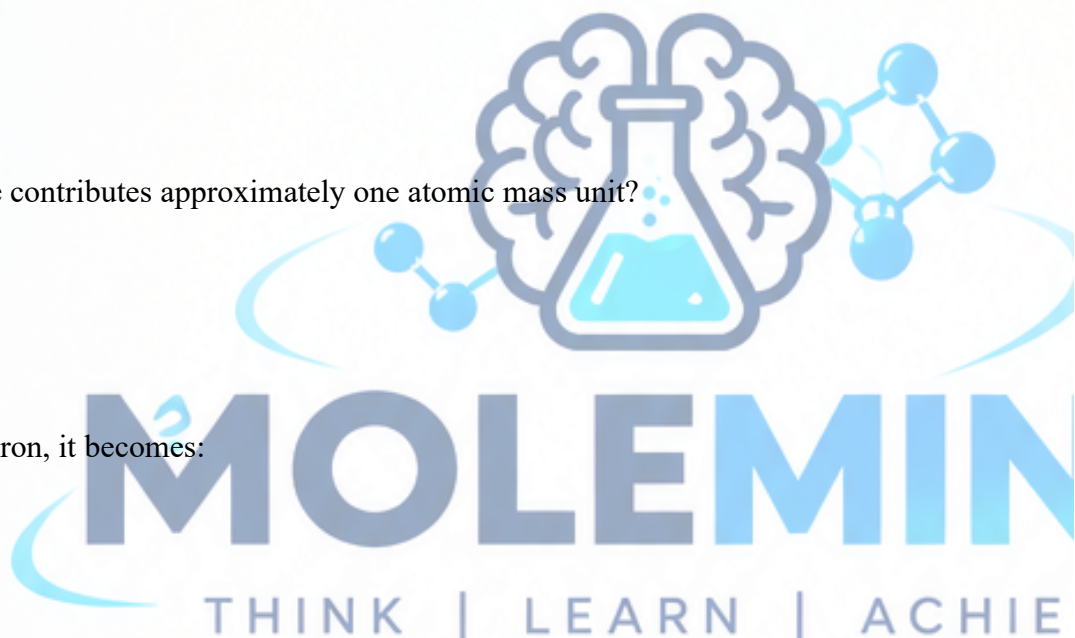
- A) Electron only
B) Proton and neutron
C) Electron and proton
D) Electron and neutron

Q114. If an atom loses one electron, it becomes:

- A) An anion
B) A cation
C) A neutron
D) An isotope

Q115. If an atom gains two electrons, its charge becomes:

- A) +2
B) +1
C) -1
D) -2



Q116. The maximum number of electrons that can be accommodated in the first shell is:

- A) 1
- B) 2
- C) 8
- D) 18

Q117. The electronic configuration of sodium (atomic number 11) is:

- A) 2, 8, 1
- B) 2, 7, 2
- C) 8, 2, 1
- D) 2, 9

Q118. An atom has electronic configuration 2,8,7. It is most likely to:

- A) Lose 7 electrons
- B) Gain 1 electron
- C) Lose 1 electron
- D) Gain 7 electrons

Q119. An element with configuration 2,8,2 will most readily form:

- A) 2- ion
- B) 1- ion
- C) 2+ ion
- D) 8+ ion

Q120. Which electronic configuration represents a noble gas?

- A) 2,8,1
- B) 2,8,7
- C) 2,8,8
- D) 2,8,6

Valency & Formulae

Q121. The valency of oxygen is generally:

- A) 1
- B) 2
- C) 3
- D) 6

Q122. The valency of aluminium is:

- A) 1
- B) 2
- C) 3
- D) 4

Q123. Formula of magnesium chloride is:

- A) MgCl
- B) Mg₂Cl

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- C) MgCl_2
D) Mg_2Cl_2

Q124. Formula of aluminium oxide is:

- A) AlO
B) Al_2O
C) AlO_2
D) Al_2O_3

Q125. Formula of calcium chloride is:

- A) CaCl
B) CaCl_2
C) Ca_2Cl
D) Ca_2Cl_2

Q126. Formula of sodium sulphide is:

- A) NaS
B) Na_2S
C) NaS_2
D) Na_2S_2

Q127. Which formula is **incorrect**?

- A) MgO
B) CaO
C) Al_2O_3
D) Na_2O_2

Q128. The correct formula of calcium hydroxide is:

- A) CaOH
B) $\text{Ca}(\text{OH})_2$
C) Ca_2OH
D) $\text{Ca}_2(\text{OH})$

Q129. In H_2O , the ratio of hydrogen atoms to oxygen atoms is:

- A) 1:1
B) 1:2
C) 2:1
D) 2:2

Q130. The total number of atoms in one molecule of H_2SO_4 is:

- A) 4
B) 5
C) 6
D) 7

Olympiad Challenge

Q131. An ion X^{2+} has 10 electrons. The atomic number of X is:

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- A) 8
B) 10
C) 12
D) 14

Q132. An ion Y^- has 18 electrons. The atomic number of Y is:

- A) 17
B) 18
C) 19
D) 16

Q133. An atom has 20 protons, 20 electrons and 20 neutrons. Which is correct?

- A) Atomic number = 20, mass number = 40
B) Atomic number = 40, mass number = 20
C) Atomic number = 20, mass number = 20
D) Atomic number = 40, mass number = 40

Q134. Which pair contains particles having the **same number of electrons**?

- A) Na^+ and Ne
B) Mg^+ and Ne
C) Cl^- and Ca^{+2}
D) Mg^{2+} and S^{2-}

Q135. An element X has configuration 2,8,8,1. Which property is most likely?

- A) It tends to gain one electron
B) It tends to lose one electron
C) It is completely unreactive
D) It tends to gain seven electrons

Q136. Which statement about noble gases is most appropriate?

- A) They have incomplete outer shells
B) They generally have stable outer electronic configurations
C) They always form +1 ions
D) They readily react with all metals

Q137. An atom has 15 electrons and 16 neutrons. Its mass number is:

- A) 15
B) 16
C) 30
D) 31

Q138. An element has atomic number 8. Which statement is correct?

- A) It has 8 neutrons necessarily
B) It has 8 protons
C) Its mass number is always 8
D) It must have 10 electrons

Q139. Which species has the greatest number of electrons?

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- A) Na^+
B) Ne
C) F^-
D) Fe^{2+}

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Q140. If X has valency 3 and Y has valency 2, the likely formula of their compound is:

- A) XY
B) X_2Y_3
C) X_3Y_2
D) X_2Y

Reasoning

Q141. A student says, "An atom with 8 protons must have a mass number of 8." The statement is:

- Isomerism
- A) Always true
B) Always false
C) Not necessarily true
D) True only for ions

Q142. Which change does **not** change the identity of an element?

- Electronic Effects
- A) Changing number of protons
B) Changing number of neutrons
C) Nuclear transformation changing protons
D) Changing atomic number

Q143. If an atom has equal numbers of protons and electrons, it is:

- Organic Chemistry
- A) Always an isotope
B) Electrically neutral
C) Always an ion
D) Negatively charged

Q144. Which one has the greatest tendency to form a positive ion?

- Concepts
→ Practice
→ Success
- A) Element with configuration 2,8,1
B) Element with configuration 2,8,7
C) Element with configuration 2,8,8
D) Element with configuration 2,6

Q145. The formula MgCl_2 indicates that:

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- A) One Mg atom combines with one Cl atom
B) One Mg atom combines with two Cl atoms
C) Two Mg atoms combine with one Cl atom
D) Two Mg atoms combine with two Cl atoms

Q146. Which pair is correctly matched?

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- A) Na^+ — anion
B) Cl^- — cation
C) Mg^{2+} — cation
D) O^{2-} — cation

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Q147. A neutral atom gains three electrons. Its charge becomes:

- A) +3
- B) +1
- C) -1
- D) -3

Q148. Which species has 10 electrons?

- A) Na^+
- B) Mg^{2+}
- C) F^-
- D) All of these

Q149. If the number of protons in an atom changes from 11 to 12, the element:

- A) Remains sodium
- B) Becomes magnesium
- C) Becomes an isotope of sodium
- D) Becomes a negative ion

Q150. An atom X has 17 protons, 18 neutrons and 18 electrons. Which statement is correct?

- A) X is a neutral chlorine atom
- B) X is Cl^- with mass number 35
- C) X is Ar^- with mass number 35
- D) X is Cl^+ with mass number 35

Chemical Reactions + Acids, Bases & Salts

Q151. Which observation most strongly indicates a chemical reaction?

- A) Change in size
- B) Formation of a new substance
- C) Change in state only
- D) Dissolving

Q152. Burning magnesium in oxygen produces:

- A) MgCl_2
- B) MgO
- C) MgCO_3
- D) Mg(OH)_2

Q153. The reaction between an acid and a base producing salt and water is called:

- A) Oxidation
- B) Reduction
- C) Neutralisation
- D) Displacement

Q154. Which gas is commonly evolved when an acid reacts with a metal such as zinc?

- A) Oxygen
- B) Nitrogen

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- C) Hydrogen
D) Carbon dioxide

Q155. Zinc + hydrochloric acid produces:

- A) Zinc oxide + hydrogen
B) Zinc chloride + hydrogen
C) Zinc hydroxide + chlorine
D) Zinc carbonate + hydrogen

Q156. Which is an example of a displacement reaction?

- A) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
B) $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$
C) $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
D) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$

Q157. In the reaction $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$, zinc is:

- A) Less reactive than copper
B) More reactive than copper
C) A non-metal
D) An acid

Q158. Which reaction is a decomposition reaction?

- A) $\text{CaCO}_3 \rightarrow \text{CaO} + \text{CO}_2$
B) $\text{Zn} + \text{CuSO}_4 \rightarrow \text{ZnSO}_4 + \text{Cu}$
C) $\text{HCl} + \text{NaOH} \rightarrow \text{NaCl} + \text{H}_2\text{O}$
D) $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$

Q159. A reaction in which two or more substances combine to form one product is:

- A) Decomposition
B) Combination
C) Displacement
D) Neutralisation

Q160. Which is an oxidation process?

- A) Addition of oxygen
B) Removal of oxygen only
C) Addition of hydrogen only
D) Loss of mass only

Acids, Bases & Indicators

Q161. Blue litmus turns red in:

- A) Base
B) Acid
C) Salt only
D) Distilled water always

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Q162. Red litmus turns blue in:

- A) Acid
- B) Base
- C) Neutral solution
- D) Sugar solution

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Q163. A solution with pH 2 is:

- A) Strongly acidic
- B) Weakly acidic
- C) Neutral
- D) Basic

Q164. A solution with pH 7 at standard conditions is:

- Isomerism
- A) Acidic
 - B) Basic
 - C) Neutral
 - D) Strongly acidic

Q165. A solution with pH 12 is:

- Electronic
Effects
- A) Acidic
 - B) Basic
 - C) Neutral
 - D) Salt-free

Q166. Which substance is commonly used as a natural indicator?

- Organic
Chemistry
- A) Turmeric
 - B) Sodium chloride
 - C) Sugar
 - D) Sand

Q167. Turmeric generally turns reddish-brown in:

- Concepts
→ Practice
→ Success
- A) Acidic solution
 - B) Basic solution
 - C) Neutral solution
 - D) Distilled water

Q168. Which substance is likely to have a pH below 7?

- A) Soap solution
- B) Baking soda solution
- C) Lemon juice
- D) Lime water

Q169. Which is generally basic?

- A) Vinegar
- B) Lemon juice
- C) Soap solution
- D) Orange juice



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Q170. An acid reacts with a carbonate to produce:

- A) Hydrogen only
- B) Oxygen only
- C) Carbon dioxide
- D) Nitrogen

Tricky Conceptual Questions

Q171. When dilute HCl is added to sodium carbonate, the gas evolved is:

- A) H_2
- B) O_2
- C) CO_2
- D) Cl_2

Q172. Carbon dioxide produced in the above reaction can be identified because it:

- A) Relights a glowing splint
- B) Turns limewater milky
- C) Burns with a pop sound
- D) Turns blue litmus blue

Q173. Antacids are used to relieve excess stomach acidity because they:

- A) Increase acidity
- B) Neutralise excess acid
- C) Produce more acid
- D) Stop digestion completely

Q174. Tooth enamel can be damaged when mouth conditions become too acidic. A basic toothpaste can help mainly by:

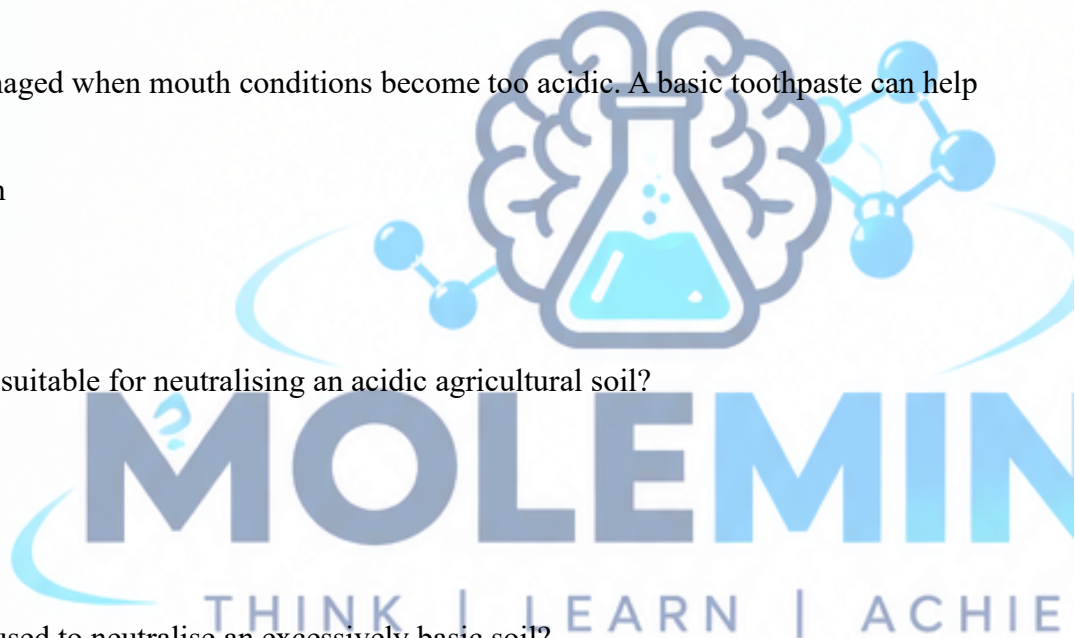
- A) Increasing acid concentration
- B) Neutralising acids
- C) Producing carbon dioxide
- D) Dissolving enamel faster

Q175. Which substance is most suitable for neutralising an acidic agricultural soil?

- A) Quicklime/lime
- B) Vinegar
- C) Lemon juice
- D) Hydrochloric acid

Q176. Which substance can be used to neutralise an excessively basic soil?

- A) Suitable organic acidic material
- B) Sodium hydroxide
- C) Lime
- D) Baking soda



Q177. A student adds phenolphthalein to solution X and it becomes pink. X is most likely:

- A) Acidic
- B) Basic
- C) Neutral
- D) Pure water

Q178. Methyl orange in an acidic solution appears:

- A) Yellow
- B) Red
- C) Blue
- D) Green

Q179. A solution turns blue litmus red but does not conduct electricity appreciably. Which conclusion is safest?

- A) It must be a strong acid
- B) It is acidic, but strength cannot be determined from litmus alone
- C) It must be neutral
- D) It must be a metal

Q180. Which statement is correct?

- A) All acids are equally strong
- B) Acidity and concentration are exactly the same concept
- C) pH gives information about acidity of an aqueous solution
- D) Every acid has pH 1

Olympiad-Level Chemistry

Q181. Three solutions have pH values 3, 7 and 11. Which statement is correct?

- A) pH 3 is basic
- B) pH 7 is acidic
- C) pH 11 is basic
- D) All are neutral

Q182. Which solution is expected to have the highest concentration of H^+ ions among the following?

- A) pH 2
- B) pH 4
- C) pH 6
- D) pH 7

Q183. A student mixes equal volumes of an acidic and a basic solution. The final solution is neutral **only if**:

- A) Both have equal volume
- B) Their neutralising capacities are appropriately balanced
- C) Both are colourless
- D) Both have pH 7 initially

Q184. A metal X reacts with dilute HCl but metal Y does not. Which conclusion is reasonable?

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- A) Y is necessarily an acid
 - B) X is above hydrogen in the relevant reactivity series
 - C) X must be less reactive than hydrogen
 - D) Y must be a noble gas

Q185. Copper does not liberate hydrogen readily from dilute HCl because:

- A) Copper is above hydrogen
- B) Copper is below hydrogen in the reactivity series
- C) HCl is not an acid
- D) Copper is a non-metal

Q186. A colourless solution gives a blue colour with blue litmus unchanged but turns phenolphthalein pink. It is most likely:

- Isomerism
- A) Basic
 - B) Acidic
 - C) Neutral
 - D) A gas

Q187. Which reaction involves **both oxidation and reduction**?

- Electronic
Effects
- A) Redox reaction
 - B) Neutralisation only
 - C) Physical change
 - D) Dissolution

Q188. In $\text{CuO} + \text{H}_2 \rightarrow \text{Cu} + \text{H}_2\text{O}$, CuO is:

- Organic
Chemistry
- A) Oxidised
 - B) Reduced
 - C) Neutralised
 - D) Sublimed

Q189. In the same reaction, hydrogen is:

- Concepts
Practice
Success
- A) Oxidised
 - B) Reduced
 - C) Neither
 - D) Converted into an element

Q190. Which change would make an acid solution **less acidic**, assuming no other reaction occurs?

- A) Adding more acid
- B) Adding water
- C) Removing water
- D) Increasing acid concentration

Grand Challenge

Q191. A student performs four experiments:

- I. $\text{Zn} + \text{dilute HCl} \rightarrow \text{gas}$
- II. $\text{Cu} + \text{dilute HCl} \rightarrow \text{no visible reaction}$



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III. $\text{NaOH} + \text{HCl} \rightarrow$ temperature rise

IV. $\text{CaCO}_3 + \text{HCl} \rightarrow$ gas

Which experiments provide evidence of chemical reactions?

- A) I only
B) I and II
C) I, III and IV
D) II and IV

Q192. A metal reacts with water producing hydrogen. Which property is most relevant in predicting this behaviour?

- A) Colour
B) Position in reactivity series
C) Melting point only
D) Density only

Q193. Which combination is correctly matched?

- A) Acid + metal carbonate $\rightarrow \text{H}_2$
B) Acid + metal $\rightarrow \text{H}_2$
C) Acid + base $\rightarrow \text{CO}_2$
D) Base + metal carbonate $\rightarrow \text{O}_2$

Q194. A solution has pH 5. Another has pH 2. Which is more acidic?

- A) pH 5
B) pH 2
C) Both equally
D) Cannot compare

Q195. A student adds a few drops of acid to a basic solution containing phenolphthalein. The pink colour disappears. This indicates:

- A) The solution has become more basic
B) Acidity has increased enough to neutralise the basic character
C) Phenolphthalein has become a salt
D) No reaction occurred

Q196. Which statement about neutralisation is most accurate?

- A) It always produces a gas
B) It generally involves acid + base forming salt and water
C) It occurs only between strong acids and strong bases
D) It is always a physical change

Q197. A student has four solutions:

- P: pH 1
 - Q: pH 6
 - R: pH 8
 - S: pH 13
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Arrange them from **most acidic to most basic**.

- A) $S \rightarrow R \rightarrow Q \rightarrow P$
B) $P \rightarrow Q \rightarrow R \rightarrow S$
C) $Q \rightarrow P \rightarrow S \rightarrow R$
D) $P \rightarrow R \rightarrow Q \rightarrow S$

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Q198. A metal X displaces copper from CuSO_4 but does not displace magnesium from MgSO_4 . Which order is possible?

- A) $\text{Mg} > \text{X} > \text{Cu}$
B) $\text{X} > \text{Mg} > \text{Cu}$
C) $\text{Cu} > \text{X} > \text{Mg}$
D) $\text{Mg} > \text{Cu} > \text{X}$

Q199. Consider:

P: All bases are alkalis.

Q: All alkalis are bases.

R: Some bases are insoluble in water.

Which statements are correct?

- A) P only
B) Q only
C) Q and R
D) P and R

Q200. An unknown solution X:

- turns blue litmus red,
- reacts with zinc producing a gas,
- reacts with sodium carbonate producing another gas,
- and reacts with sodium hydroxide producing a salt.

Which conclusion is **most appropriate**?

- A) X is a base
B) X is an acid
C) X is a neutral salt
D) X is necessarily concentrated HCl

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ANSWER KEY

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	A	21	C	31	B	41	B
2	C	12	B	22	C	32	D	42	B
3	B	13	C	23	B	33	B	43	A
4	C	14	B	24	C	34	C	44	A
5	C	15	B	25	B	35	A	45	C
6	B	16	B	26	A	36	B	46	C
7	B	17	C	27	B	37	B	47	B
8	B	18	B	28	B	38	B	48	A
9	C	19	A	29	A	39	B	49	B
10	C	20	C	30	C	40	B	50	C

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
51	B	61	B	71	B	81	C	91	A
52	C	62	B	72	A	82	B	92	D
53	C	63	B	73	B	83	B	93	A
54	D	64	C	74	B	84	B	94	B
55	B	65	B	75	B	85	B	95	B
56	C	66	B	76	A	86	B	96	B
57	A	67	C	77	B	87	B	97	B
58	A	68	B	78	A	88	B	98	A
59	C	69	B	79	B	89	B	99	B
60	C	70	A	80	B	90	C	100	A

Q	Ans	Q	Ans	Q	Ans	Q	Ans
101	B	126	B	151	B	176	A
102	C	127	D	152	B	177	B
103	C	128	B	153	C	178	B
104	B	129	C	154	C	179	B
105	C	130	D	155	B	180	C
106	B	131	C	156	B	181	C
107	B	132	A	157	B	182	A
108	B	133	A	158	A	183	B
109	B	134	A	159	B	184	B
110	B	135	B	160	A	185	B
111	A	136	B	161	B	186	A
112	C	137	D	162	B	187	A
113	B	138	B	163	A	188	B
114	B	139	D	164	C	189	A
115	D	140	B	165	B	190	B
116	B	141	C	166	A	191	C
117	A	142	B	167	B	192	B
118	B	143	B	168	C	193	B
119	C	144	A	169	C	194	B
120	C	145	B	170	C	195	B
121	B	146	C	171	C	196	B
122	C	147	D	172	B	197	B
123	C	148	D	173	B	198	A
124	D	149	B	174	B	199	C
125	B	150	B	175	A	200	B

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Isomerism



Electronic
Effects



Organic
Chemistry



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